THE LABOR MARKET
CHAPTER 7. BLANCHARD

MACROECONOMICS

PAULINA ETXEBERRIA GARAIGORTA

Reinventing Higher Education

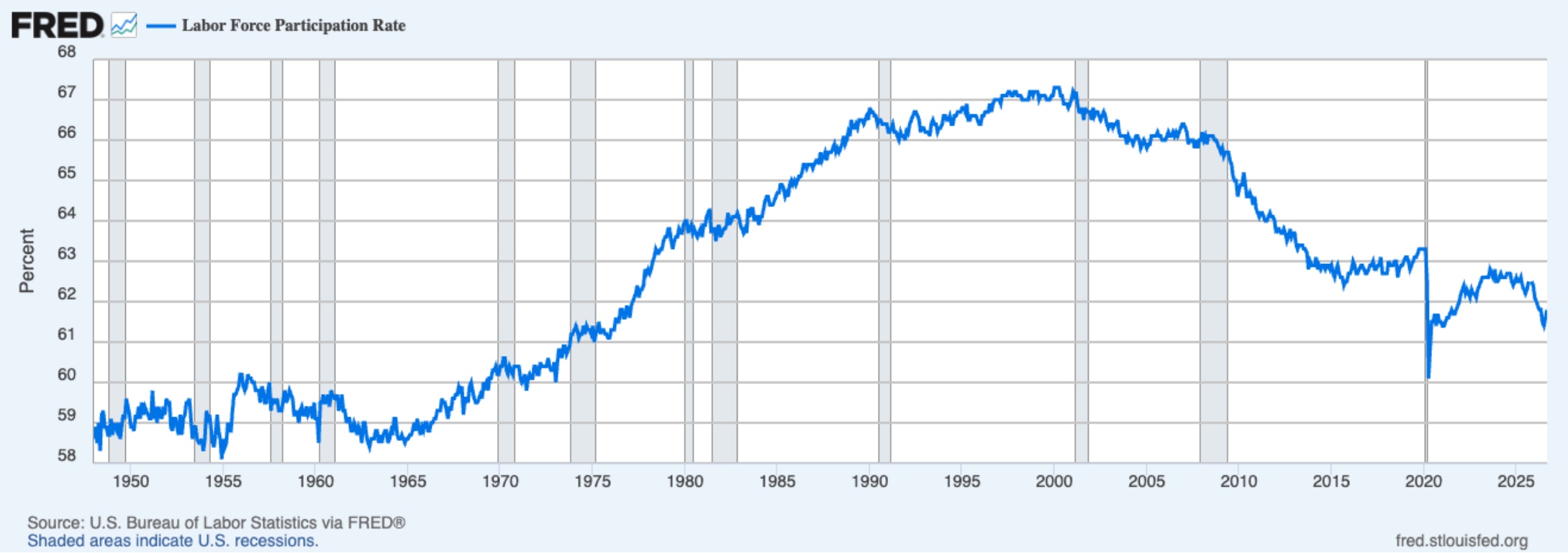
Why do governments devote a significant amount of resources to measuring unemployment?

ENORMOUS ECONOMIC & SOCIAL COSTS
FOR
INDIVIDUALS, FAMILIES AND THE ECONOMY.

Unemployment Rate in the US 1948-2026



LABOR FORCE PARTICIPATION RATE: US 1948-2026



1. A Tour of the Labor Market

Potential workers
= Non-institutional civilian population

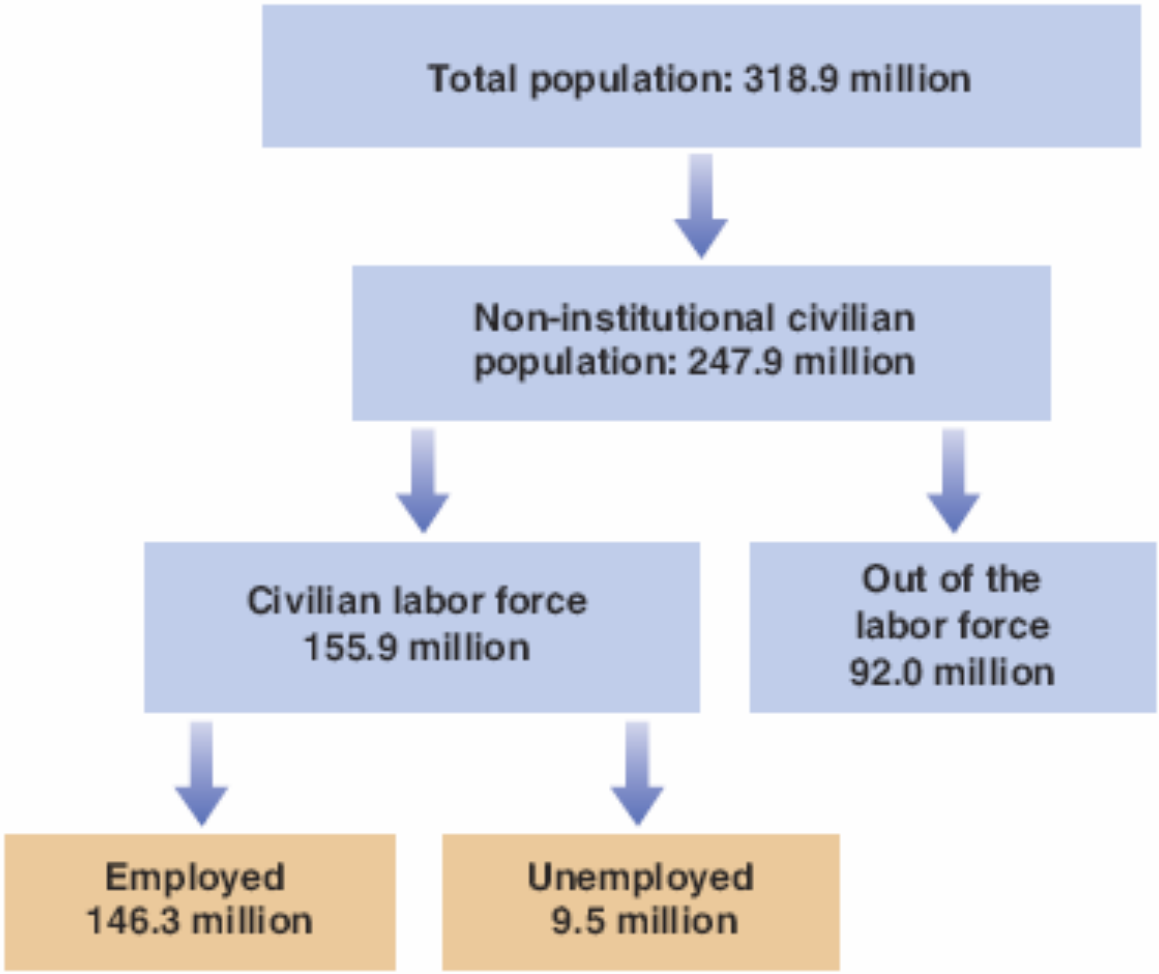
Everyone in the general population except:

- Children under 16 years
- Active military personnel
- Institutionalized persons

Unemployment rate= 9.5/155.9 = 6.1%

Source: Current Population Survey http://www.bls.gov/cps/.

Figure 7-1 Population, Labor Force, Employment, and Unemployment in the US (in millions), 2014



Exercise 2:
Calculate the labor force participation & unemployment rates

United States (in millions)	Sep-08	Sep-14
National Population	305.9	318.2
Children + active military + institutionalized persons	71.5	69.8
Potential workers	234.4	248.4
Not in Labor force	79.8	92.6
Labor force	154.6	155.8
Employed persons	145.1	146.6
Unemployed persons	9.5	9.3
Labor force population rate =	___._%	___._%
Unemployment rate =	___._%	___._%

Source: Bureau of Labor Statistics.

Solution:

United States (in millions)	Sep-08	Sep-14
National Population	305.9	318.2
Children + active military + institutionalized persons	71.5	69.8
Potential workers	234.4	248.4
Not in Labor force	79.8	92.6
Labor force	154.6	155.8
Employed persons	145.1	146.6
Unemployed persons	9.5	9.3
Labor force population rate =	**66.0%**	**62.7%**
Unemployment rate =	**6.1%**	**6.0%**

The labor force participation rate has dropped due to the increase in potential workers not in the labor force. As a result, the labor force and employment have remained essentially the same, and therefore the unemployment rate has fallen back to 6%.

https://www.cbsnews.com/news/the-economys-hidden-problem-working-class-men-without-jobs/

Exercise 3:

What is the classification of each of these people?

- Unionized electrician
- Part-time coffee-shop barista
- Former auto worker seeking new employment
- Former auto worker collecting disability
- Full-time college student
- Retired grandmother
- Stay-at-home father

Exercise 4:
Compare Economies A & B

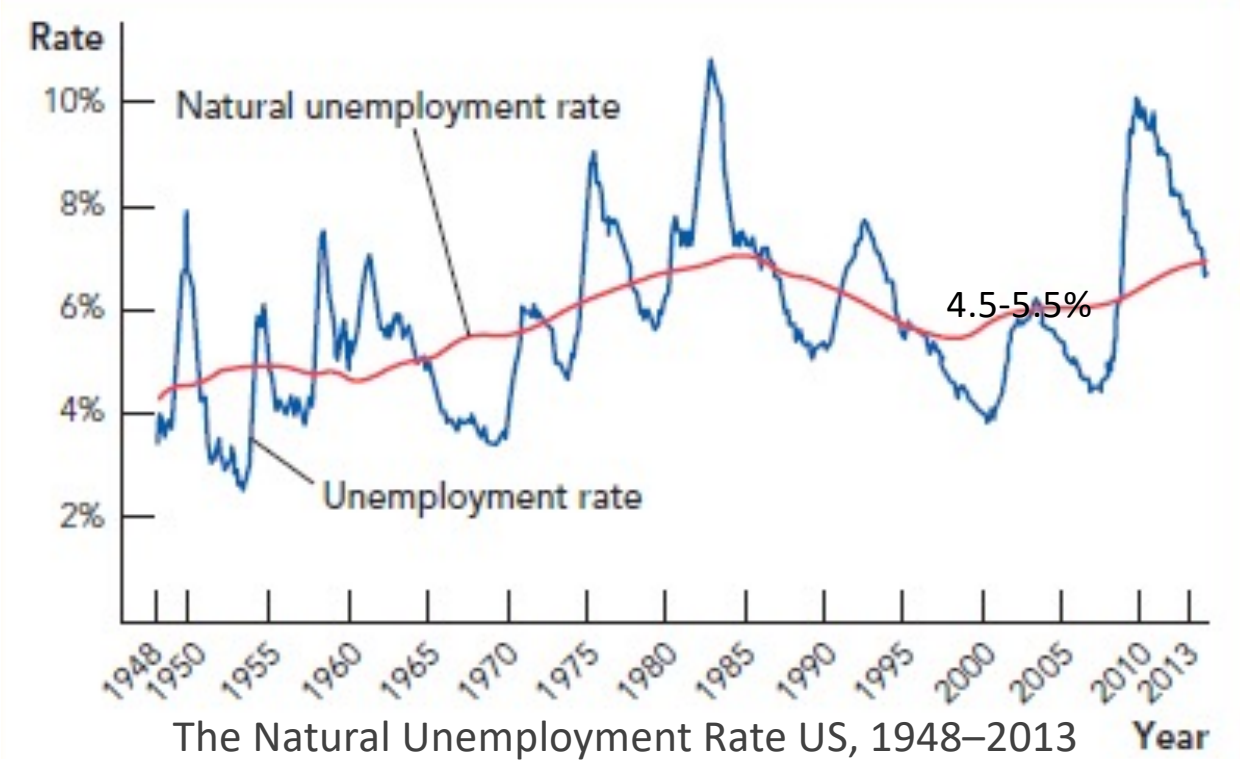
What is the Unemployment rate in both Economies?

Are the conditions the same in both labor markets?

- The Labor Force in two similar economies (Economies A and B) consists of 10 people: 8 employed and 2 unemployed people.
- The unemployment rate is 20 percent in both economies.
- In Economy A, one of the two unemployed people finds a job and becomes employed.
- In Economy B, one of the two unemployed people gave up hope of finding a job and stops seeking employment.

3. Five Different Kinds of Unemployment

$$actual\ unemployment = \boxed{frictional + structural} + cyclical$$



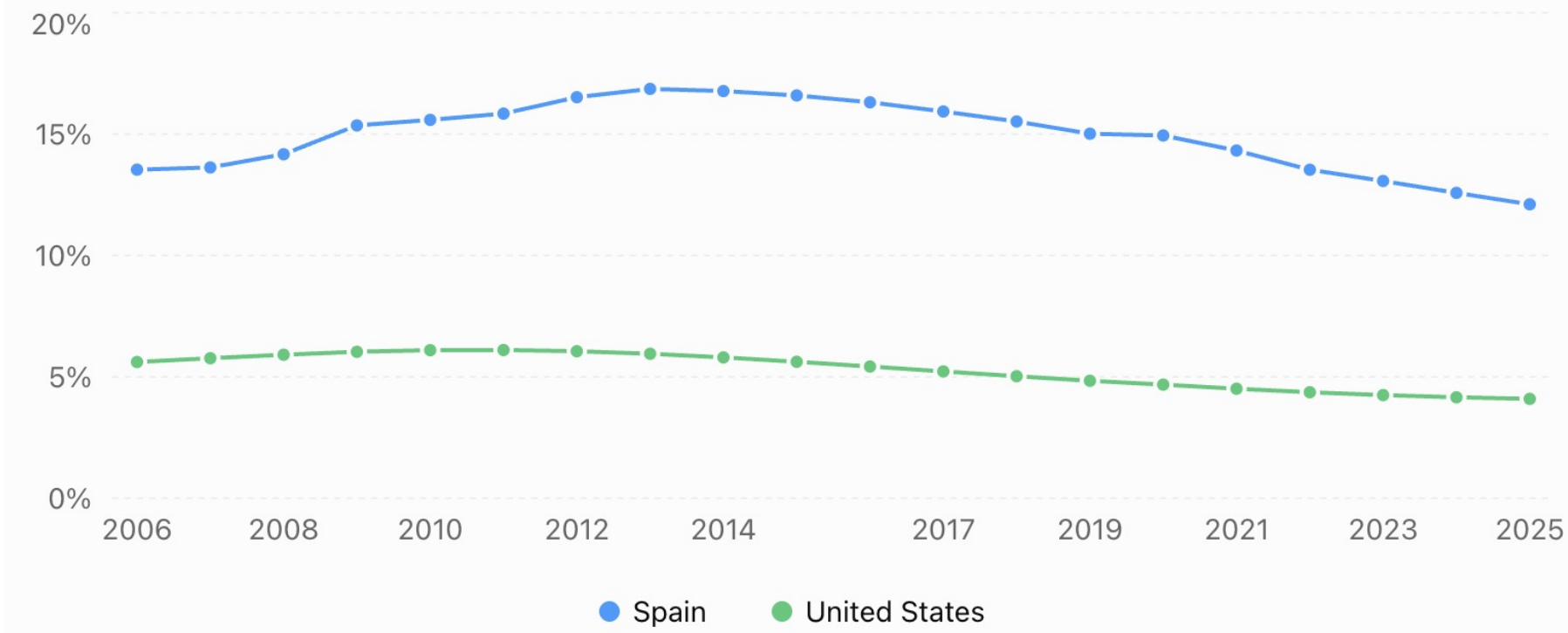
The Natural Unemployment Rate US, 1948–2013

Natural Rate of Unemployment

- **Spain** averaged **16.44%:** 1976-2021
- The Bank of Spain estimates a **structural unemployment** rate of **11.4% of the active population for 2027**
- All time high of 26.94% in 2013Q1

Estimated NAWRU: Spain vs United States, 2006–2025

Annual NAWRU estimates, used as a proxy for the natural rate of unemployment.



Source: European Commission AMECO series. Estimates are model-based and revisable.

ECONOMIC DATA | SINCE 1991

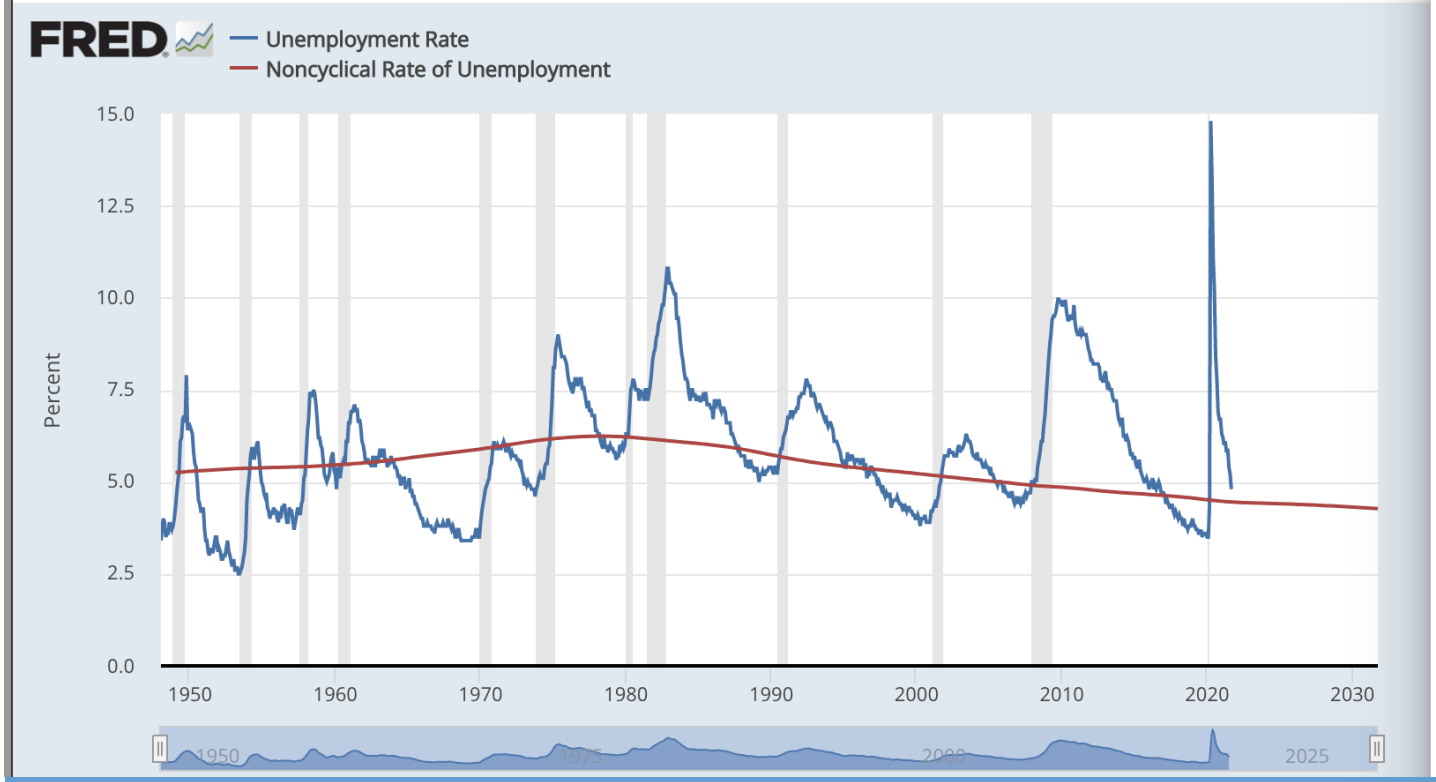
Release Calendar FRED Too

Categories > Population, Employment, & Labor Markets > Current Population Survey (Household Survey) > Unemployment Rate

FRED Graph

https://fred.stlouisfed.org/series/UNRATE#0

1Y | 5Y | 10Y | Max



EDIT LINES 2 | ADD LINE | FORMAT

EDIT LINE 2

(a) Noncyclical Rate of Unemployment, Percent, Not Seasonally Adjusted **(NROU)**

Units:

Percent — Copy to all

Modify frequency:

Quarterly

Customize data:
Write a custom formula to transform one or more series or combine two or more series.

You can begin by adding a series to combine with your existing series.

Select... — Add

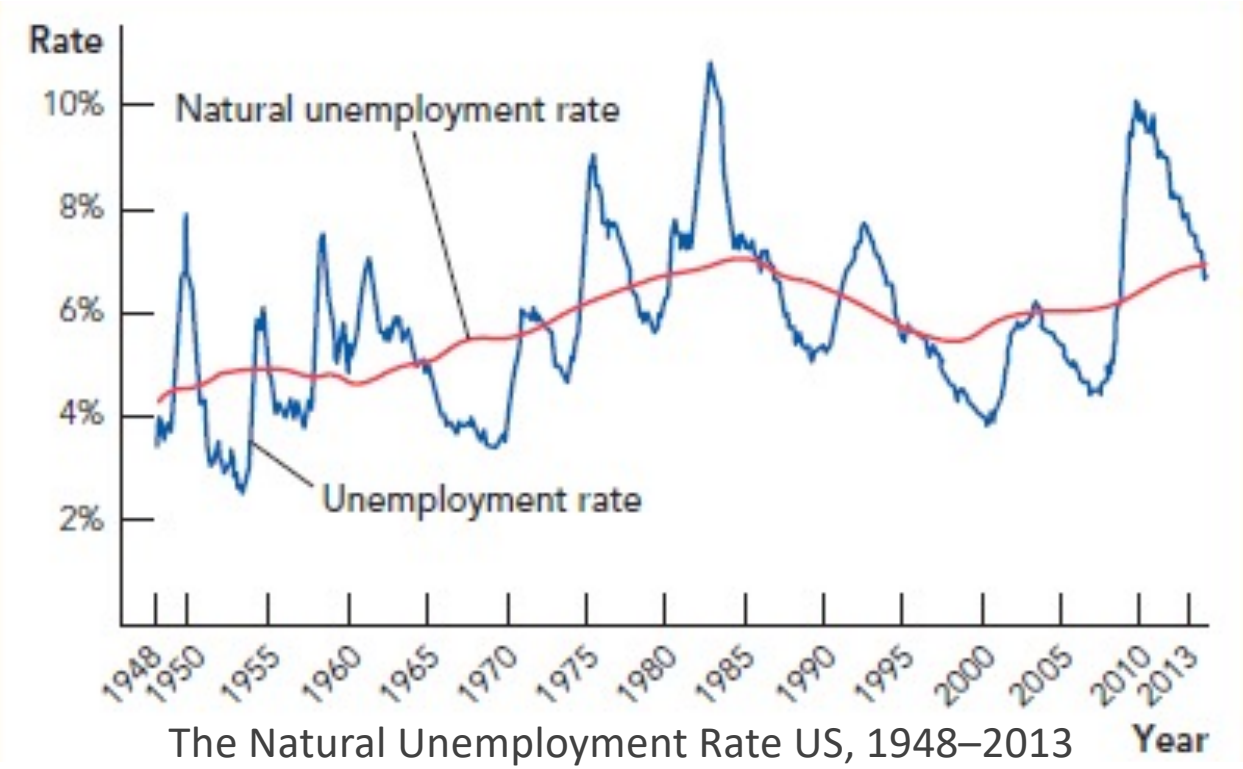
Now create a custom formula to combine or transform the series.
Need help? [+]

Formula: a — Apply

Finally, you can change the units of your new series.

3. Five Different Kinds of Unemployment

$$actual\ unemployment = \boxed{frictional + structural} + cyclical$$



The Natural Unemployment Rate US, 1948–2013

Natural Rate of Unemployment

3. Five Different Kinds of Unemployment

Exercise: One cause of structural unemployment is:

a. A recession.
b. A large, economy-wide shock.
c. The time it takes to find a suitable job.

Answer: b – *one cause of structural unemployment is a large, economy-wide shock.*

3. Five Different Kinds of Unemployment

Exercise: Cyclical unemployment is correlated with:

a. Innovation.
b. The business cycle.
c. Economic shocks.

Answer: b – *cyclical unemployment is correlated with the business cycle.*

3. Five Different Kinds of Unemployment

1. **Actual Unemployment**
2. **Cyclical unemployment:** Associated with short-run fluctuations in output
3. **The Natural rate of Unemployment:** Rate that would prevail with no cyclical unemployment

4. **Frictional unemployment**: workers between jobs in the dynamic economy

5. **Structural unemployment**: labor market failure to match workers and firms

3. Five Different Kinds of Unemployment

2. Cyclical Unemployment

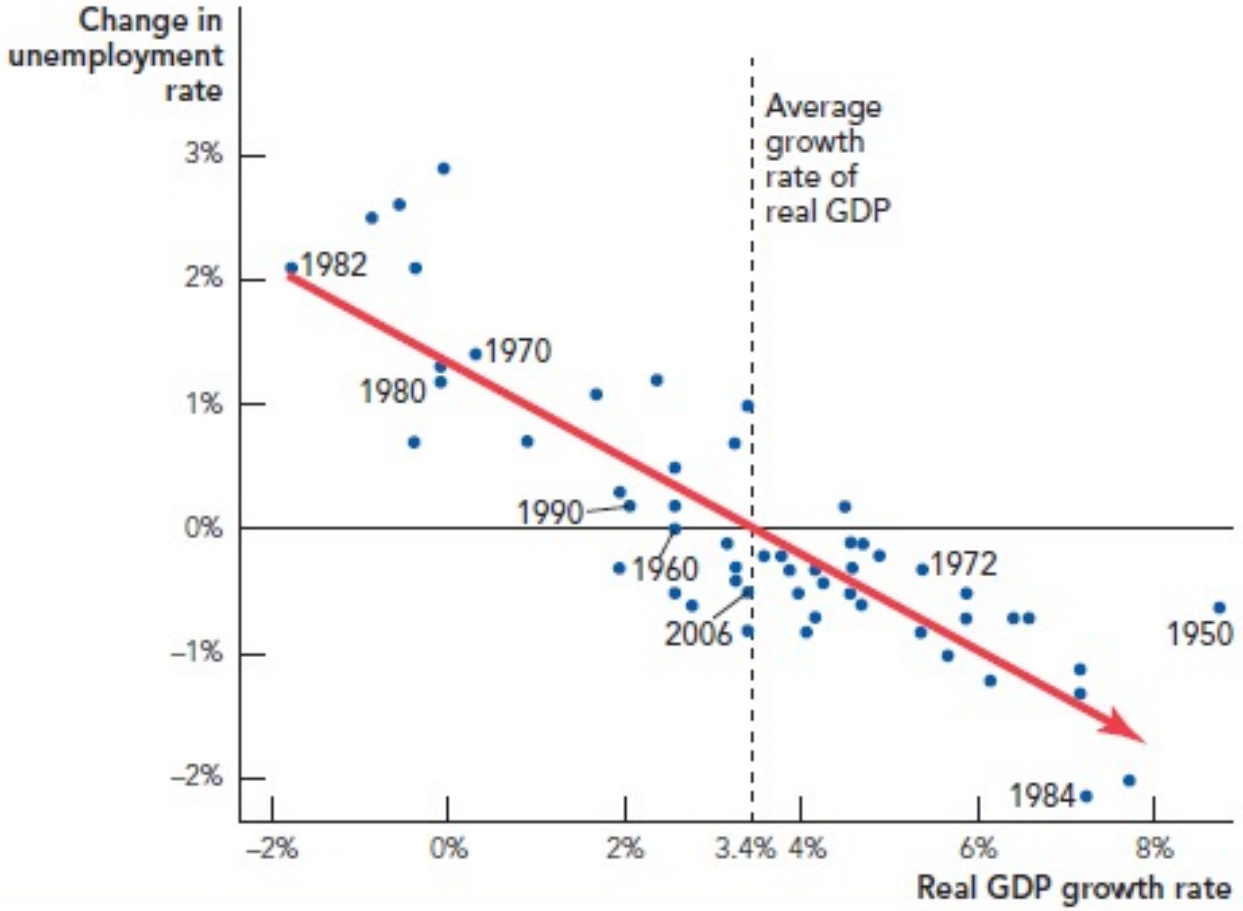
- Is correlated with the ups and downs of the business cycle.
- The difference between the actual rate and the natural rate.
- Cyclical employment can increase or decrease dramatically over a matter of months for two reasons:

1. When GDP is falling, firms often lay off workers, which increases unemployment.
2. When workers are idle, it's likely that related capital is also idle and the economy cannot be maximizing growth.

3. Five Different Kinds of Unemployment

Cyclical Unemployment

Faster Growth in Real GDP Decreases Unemployment



3. Five Different Kinds of Unemployment

3. The Natural Rate of Unemployment

- The natural rate of unemployment is what would prevail if the economy were not in a recession or expansion.
- These medium-run changes can be divided into two kinds:
  - Frictional
  - Structural
- The underlying rates of frictional and structural unemployment change only slowly through time as major, long-lasting features of the economy change.
- The natural rate changes only slowly through time and the actual rate of unemployment varies around the natural rate.

3. Five Different Kinds of Unemployment

4. Frictional Unemployment

- Results when workers are changing jobs in a dynamic economy.
- Short term unemployment caused by the ordinary difficulties of matching employee to employer.
- Jobs are constantly being created and destroyed.
- The process of seeking a new job could result in temporary unemployment.
- Some amount of frictional unemployment is efficient.
- Instead of accepting the first available job, workers take time to match their skills with demands of firms.

3. Five Different Kinds of Unemployment

4. Frictional Unemployment

- Finding a job that you want at a wage that you will accept, and the employer will pay takes time.
- This leads to frictional unemployment.
- Scarcity of information is one of the causes of frictional unemployment.

- In the United States, most of the natural rate of unemployment is frictional.
  - The labor market operates relative smoothly.
  - The same is not necessarily true in Europe.

3. Five Different Kinds of Unemployment

5. Structural Unemployment

- Persistent, long-term unemployment caused by long-lasting shocks or permanent features of an economy that make it more difficult for some workers to find jobs.
- Possible causes:
  - labor market institutions: unemployment benefits, the level of minimum wage, hiring and firing costs
  - restructuring as a result of globalization,
  - new information technologies, **AI**
  - the shift from manufacturing to services.

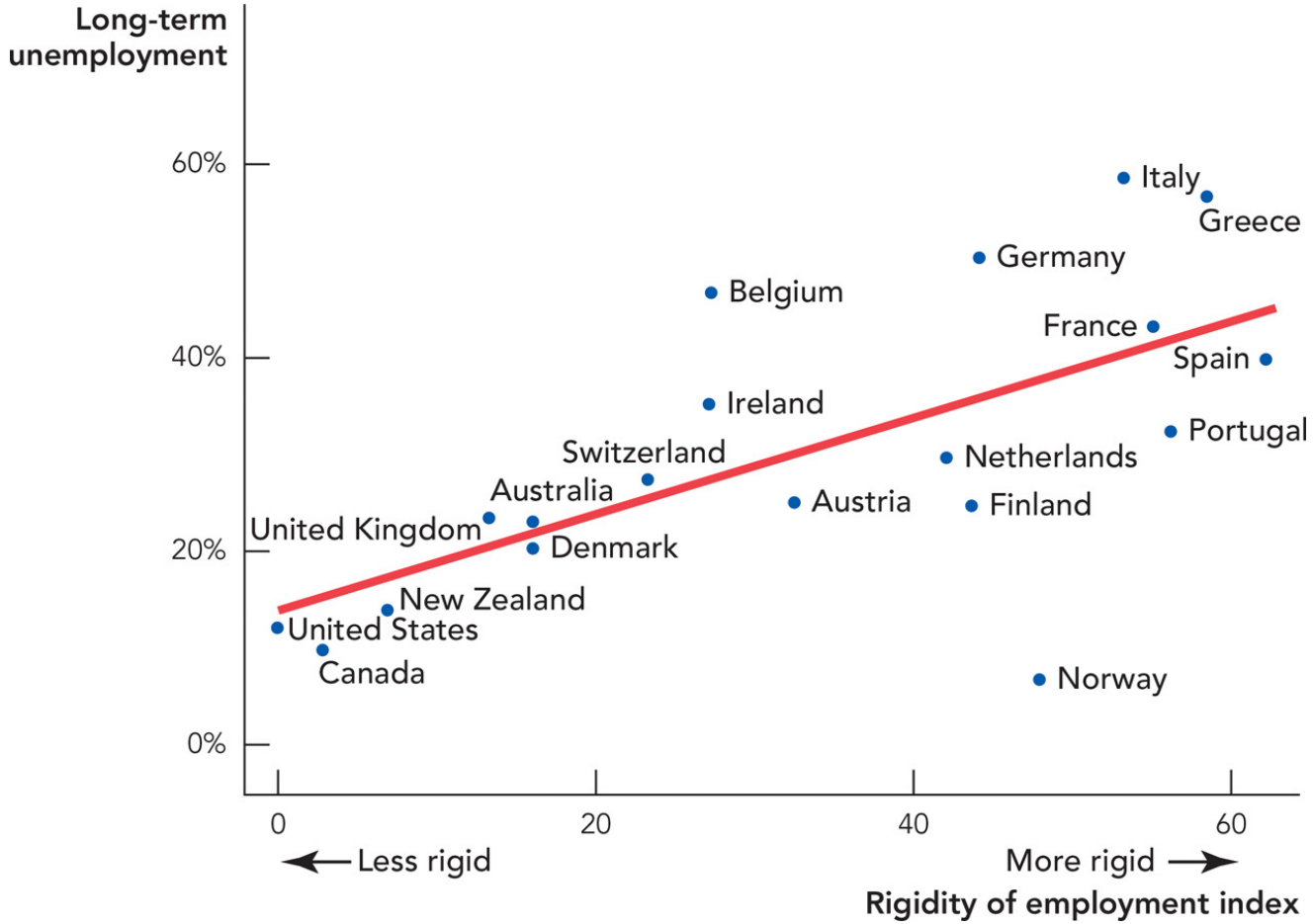
3. Five Different Kinds of Unemployment

5. Structural Unemployment

- It has significant costs:
  - Loss of economic output.
  - The unemployed suffer higher levels of stress, higher rates of suicide, and lower rates of measured happiness.
- **Structural unemployment has been a more serious problem in Europe than in the US** because of labor regulations:
  - Unemployment benefits, minimum wages, unions, and employment protection laws benefit workers.
  - All of these regulations are more generous and wide-ranging in Europe.
  - These regulations can increase unemployment rates.

3. Five Different Kinds of Unemployment

Hiring and Firing Costs Increase Long-Term Unemployment



Economy (World bank, 2025)	Unemployment (% of the labor force)	Labor force participation (% of population aged 15+)
United States	4.2%	61.7%
China	4.6%	64.6%
Germany	3.7%	60.6%
India	4.2%	55.7%
Japan	2.5%	63.4%
United Kingdom	4.7%	61.4%
France	7.5%	55.3%
Italy	6.4%	49.3%
Russia	2.1%	61.1%
Brazil	6.0%	63.1%
Canada	6.9%	64.5%
South Korea	2.7%	64.4%
Mexico	2.7%	61.6%
Spain	10.4%	57.4%

THEORETICAL EXERCISE

Some people have argued that high unemployment in some European countries relative to the US during the 1990s and 2000s was due to the existence of rigid labor market institutions (for example, powerful unions, generous unemployment benefits, and strong employment protection legislation).

1. Using Figure 16.1, check if the unemployment rate has always been higher in most European countries compared to the US.
2. Looking at Figures 16.1, 16.14, and 16.16, evaluate the claim that high unemployment in Europe was due to the existence of rigid labor market institutions.

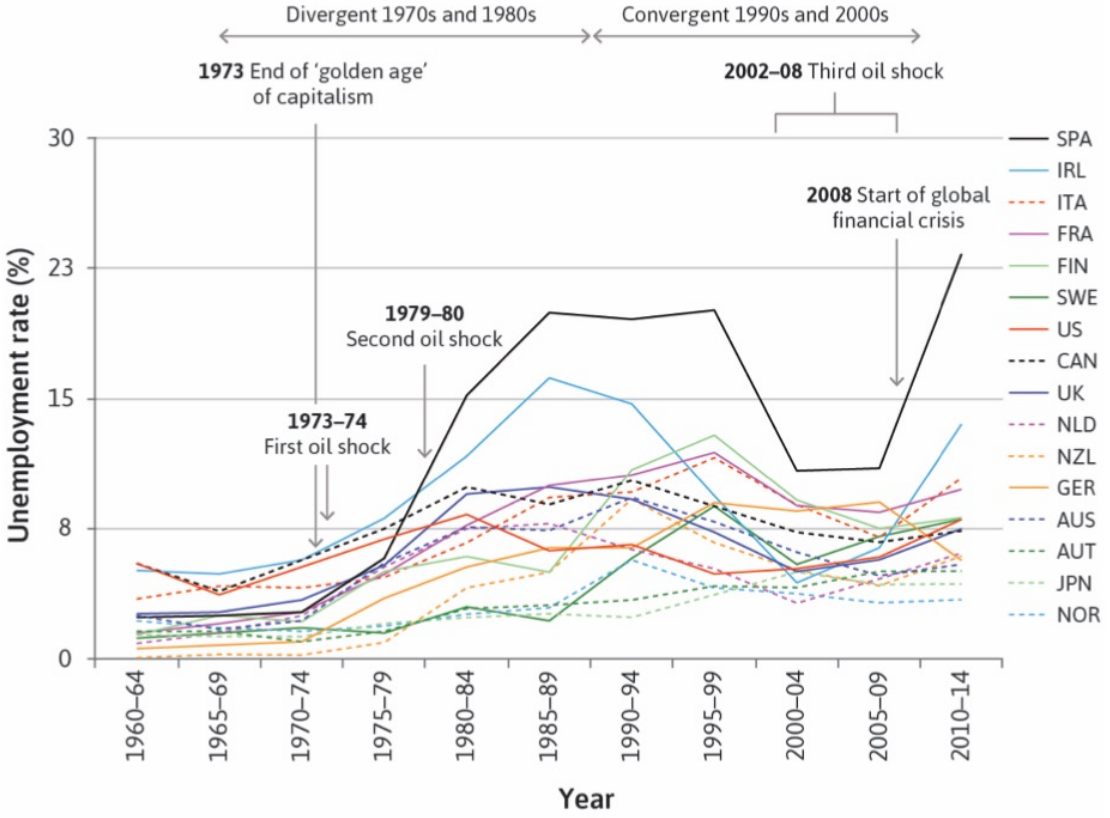


Figure 16.1 Unemployment rates in selected OECD countries (1960–2014)

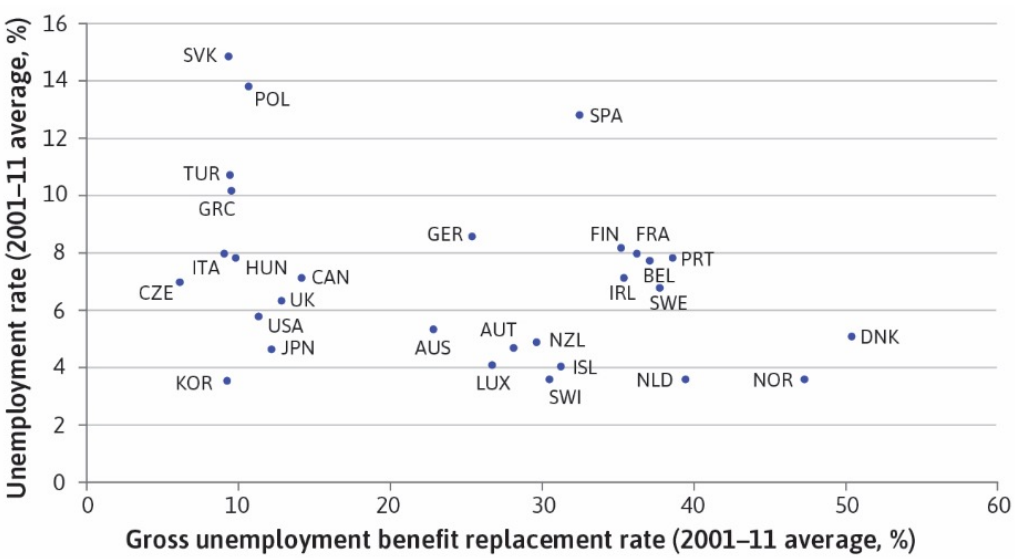


Figure 16.16 Unemployment benefit generosity and unemployment rates across the OECD (2001–2011).

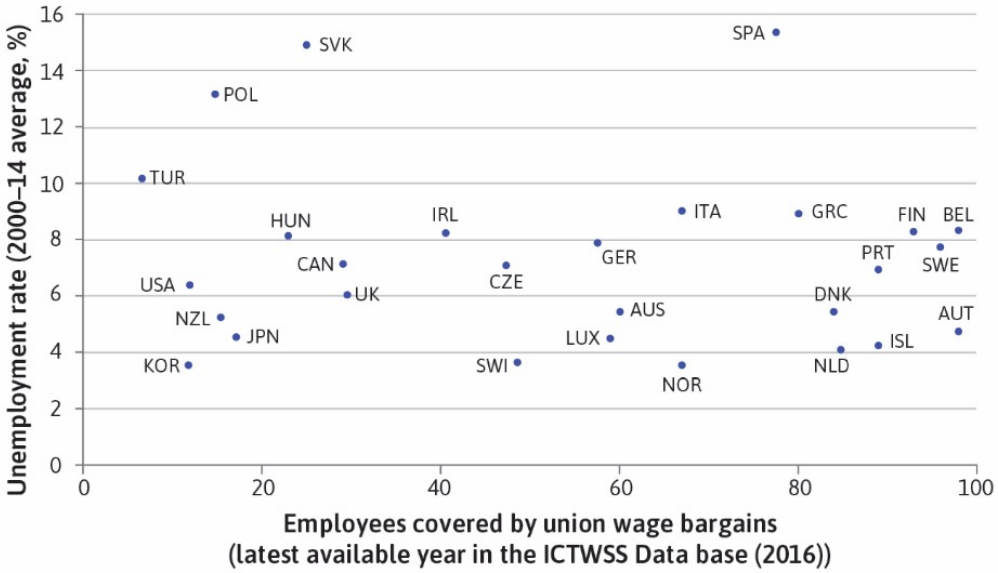


Figure 16.14 Union wage bargaining coverage and unemployment across the OECD (2000–2014).

Answer

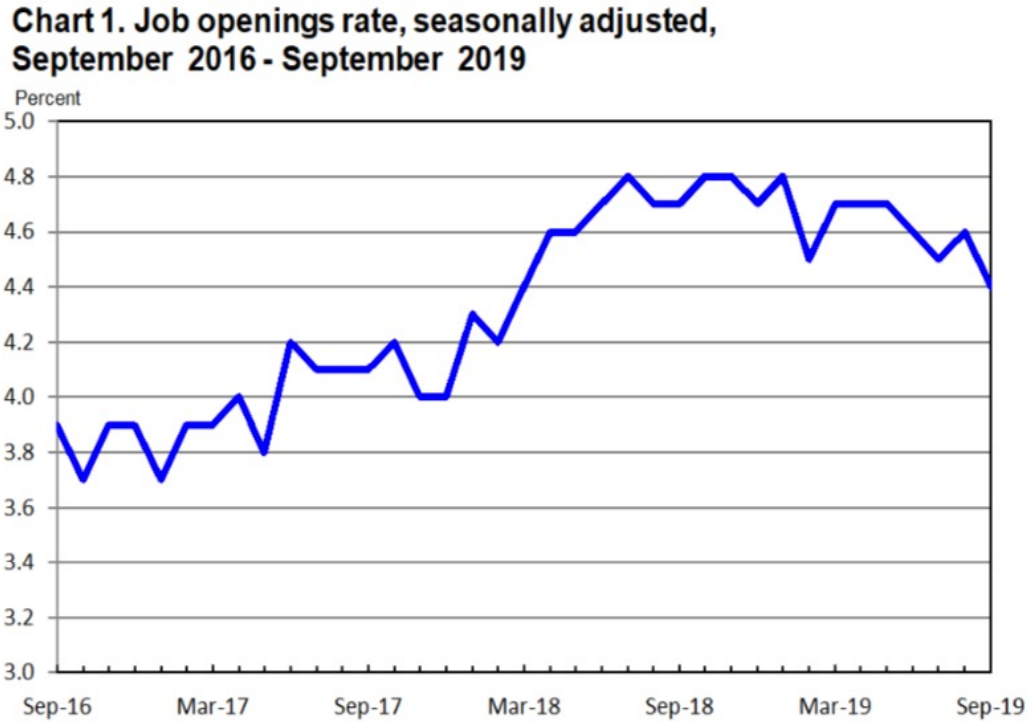
1. The US had higher unemployment rates than many European countries between 1960 and 1980. Since then, most European countries have tended to suffer unemployment rates higher than those in the US, the only European countries with rates below those of the US in this later period have been Austria, Norway, and the Netherlands.

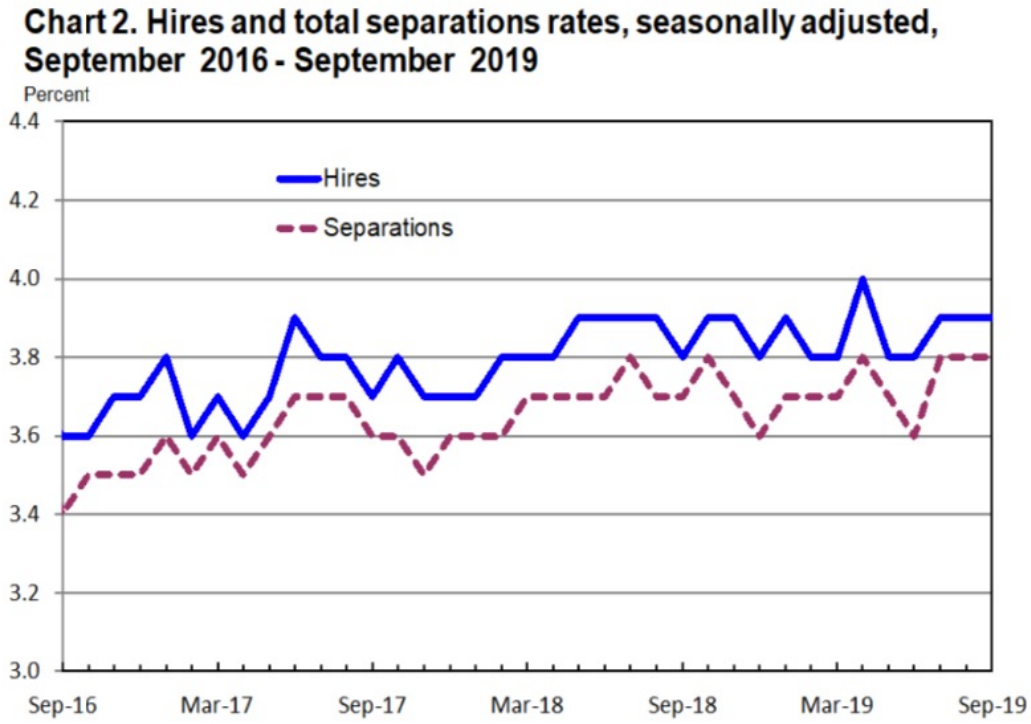
2. If rigid labor market institutions were the reason for high unemployment rates in Europe, we would like to see a **positive relationship** in Figures 16.14 and 16.16 i.e. the higher the union wage bargaining coverage and unemployment generosity, the higher the unemployment rate. Such a relationship cannot be clearly seen in these figures.

In Figure 16.14, for example, the USA has a very small fraction of employees covered by union wage bargains (around 15%), amongst the lowest in the sample. But its average unemployment rate (around 6.4%) is close to the middle of the sample. Similarly, Figure 16.16 shows the US towards the bottom end of the sample when it comes to its unemployment benefit replacement ratio (around 12%), while its unemployment rate is near the middle of the sample values (around 6.4%).

1. A Tour of the Labor Market

Hires and Separations



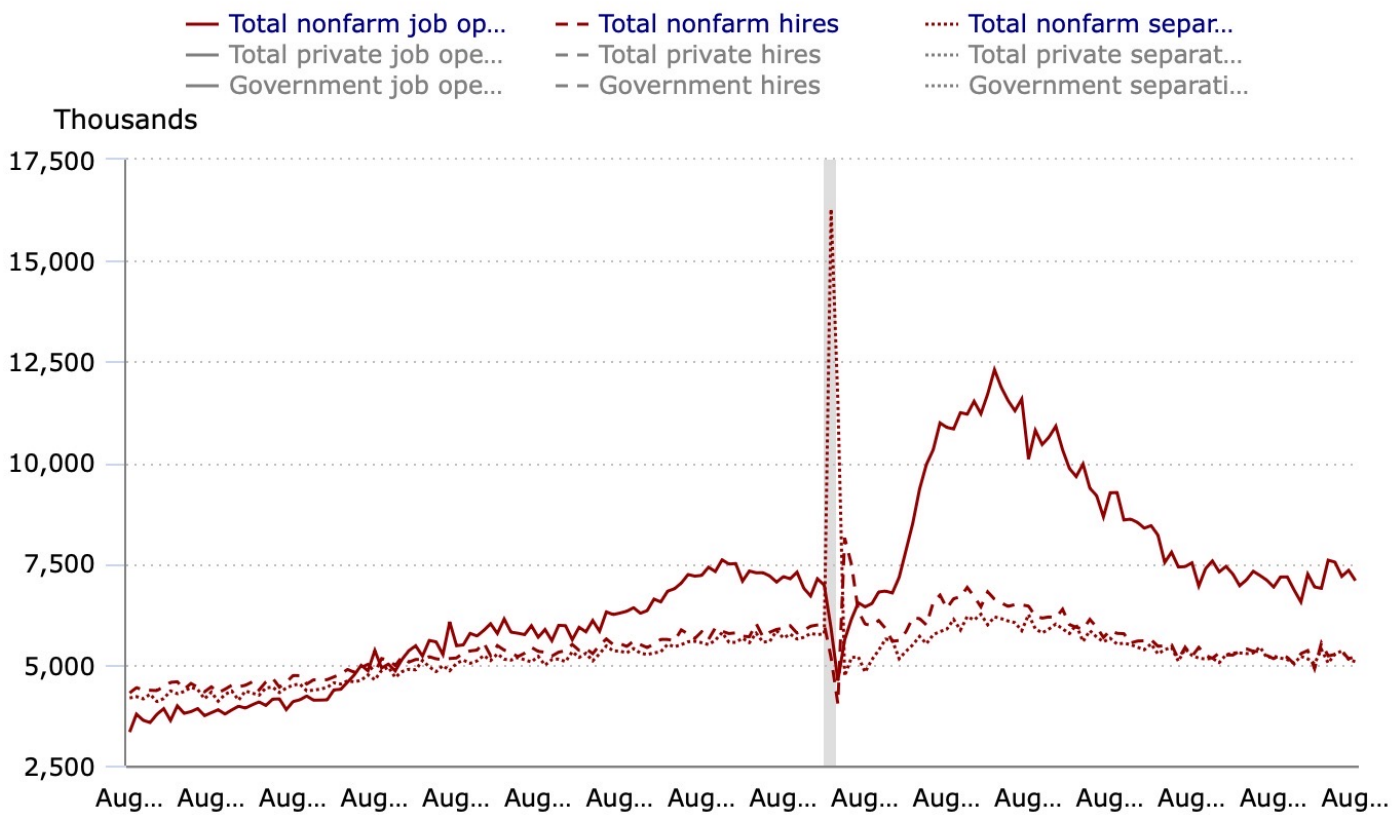


Source: BLS

https://www.bls.gov/charts/job-openings-and-labor-turnover/opening-hire-seps-level.htm

Job openings, hires, and separations levels, seasonally adjusted

Click and drag within the chart to zoom in on time periods



Source: U.S. Bureau of Labor Statistics.

1. A Tour of the Labor Market

1. The flows of workers in and out of employment are large.
2. The flows in and out of unemployment are large relative to the number of unemployed.
3. There are also large flows in and out of the labor force, much of it directly to and from employment.

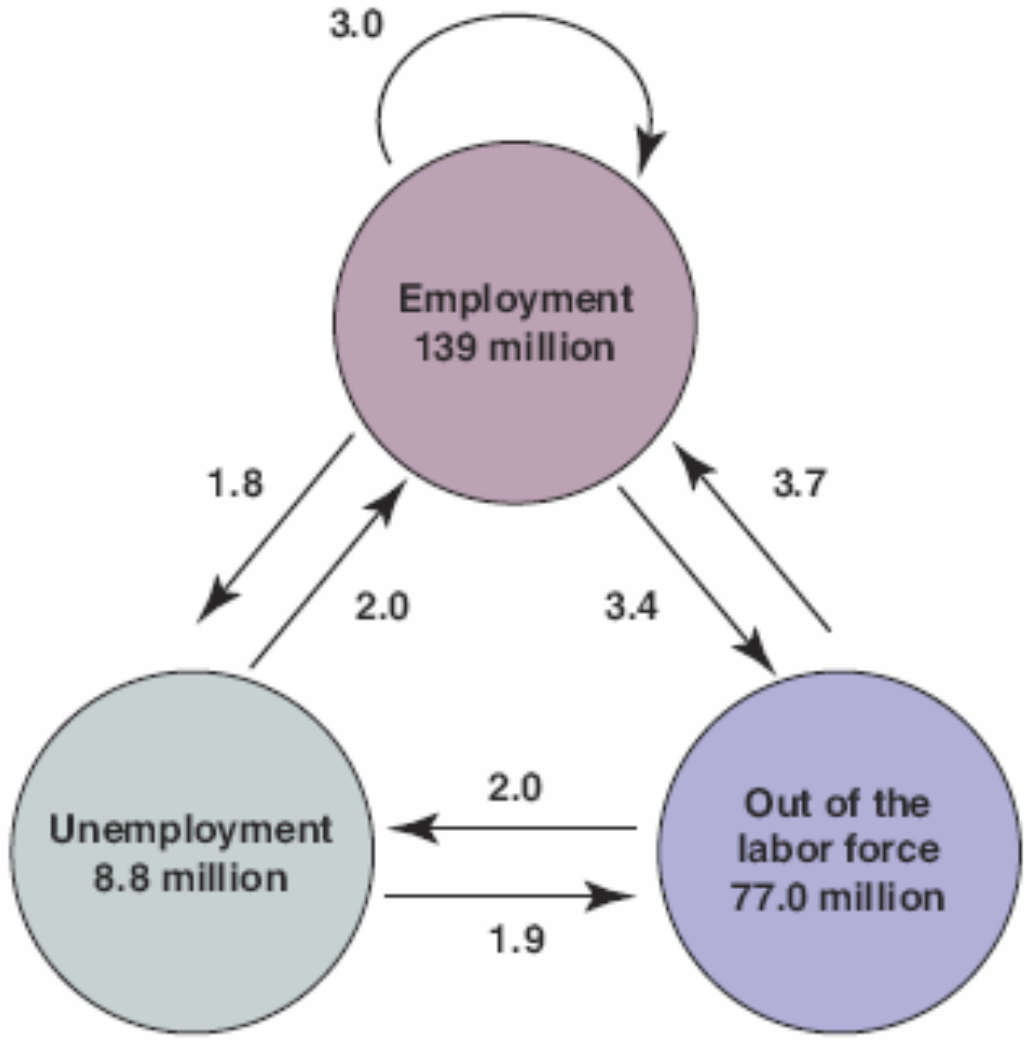


Figure 7-2 Average Monthly Flows between Employment, Unemployment, and Nonparticipation in the US, 1996 to 2014 (millions)

Source: Calculated from the series constructed by Fleischman and Fallick, http://www.federalreserve.gov/econresdata/researchdata/feds200434.xls.

1. A Tour of the Labor Market

Many who are classified as “out of the labor force” are in fact **discouraged workers**—not actively looking for a job but will take it if they find one.

So, rather than the unemployment rate, economists sometimes focus on the **Employment rate**—the ratio of employment to the population.

Employment rate = employed people ÷ DIFERENT VALUES × 100

Source: World Bank / ILO modelled estimate, indicator SL.EMP.TOTL.SP.ZS; 2025 values rounded to whole percentages.

The employment rate depends on the denominator

Common formula: employed people ÷ population in the selected age group × 100

EMPLOYMENT RATE / EMPLOYMENT–POPULATION RATIO

Employed people in the age group / Total population in the same age group

AGE 15 AND OVER

Employed, 15+
÷ population, 15+

Includes everyone aged 15+. Results reflect population age structure.

Source: World Bank WDI / ILO

AGES 15–64

Employed, 15–64
÷ population, 15–64

Conventional working-age measure, excluding under-15s and over-64s.

Source: OECD

AGES 20–64

Employed, 20–64
÷ population, 20–64

EU headline indicator; excludes typical study and retirement ages.

Source: Eurostat

BLS: AGE 16 AND OVER

Employed, 16+
÷ civilian noninstitutional population, 16+

U.S. BLS measure. Excludes active-duty military and institutional residents.

Source: U.S. BLS CPS

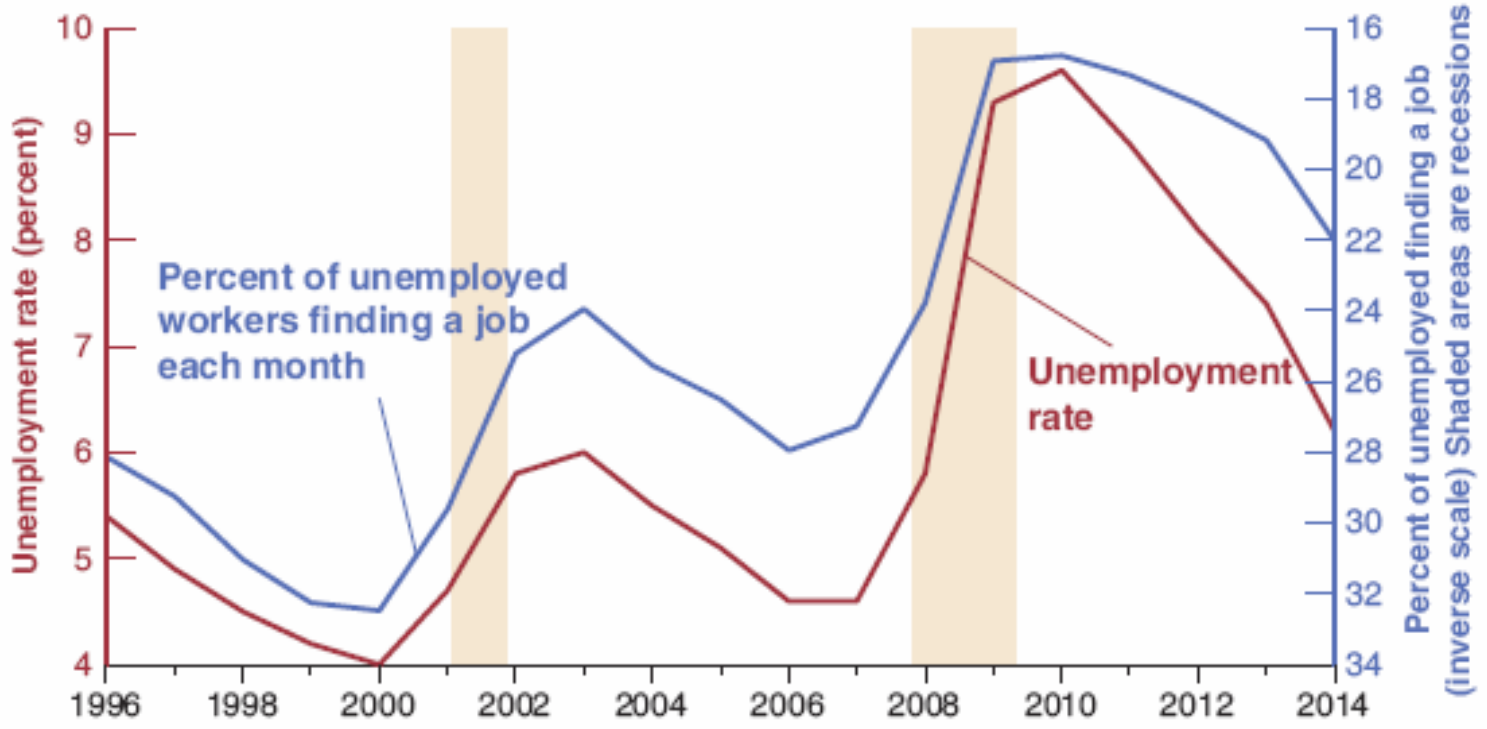
2. Movements in Unemployment

When unemployment is high, workers are worse off in two ways:

- Employed workers face a **higher probability of losing their job.**
- Unemployed workers face a **lower probability of finding a job**; or they can expect to remain unemployed for a longer time.

2. Movements in Unemployment

Figure 7-4 The Unemployment Rate and the Proportion of Unemployed Finding Jobs, 1996–2014

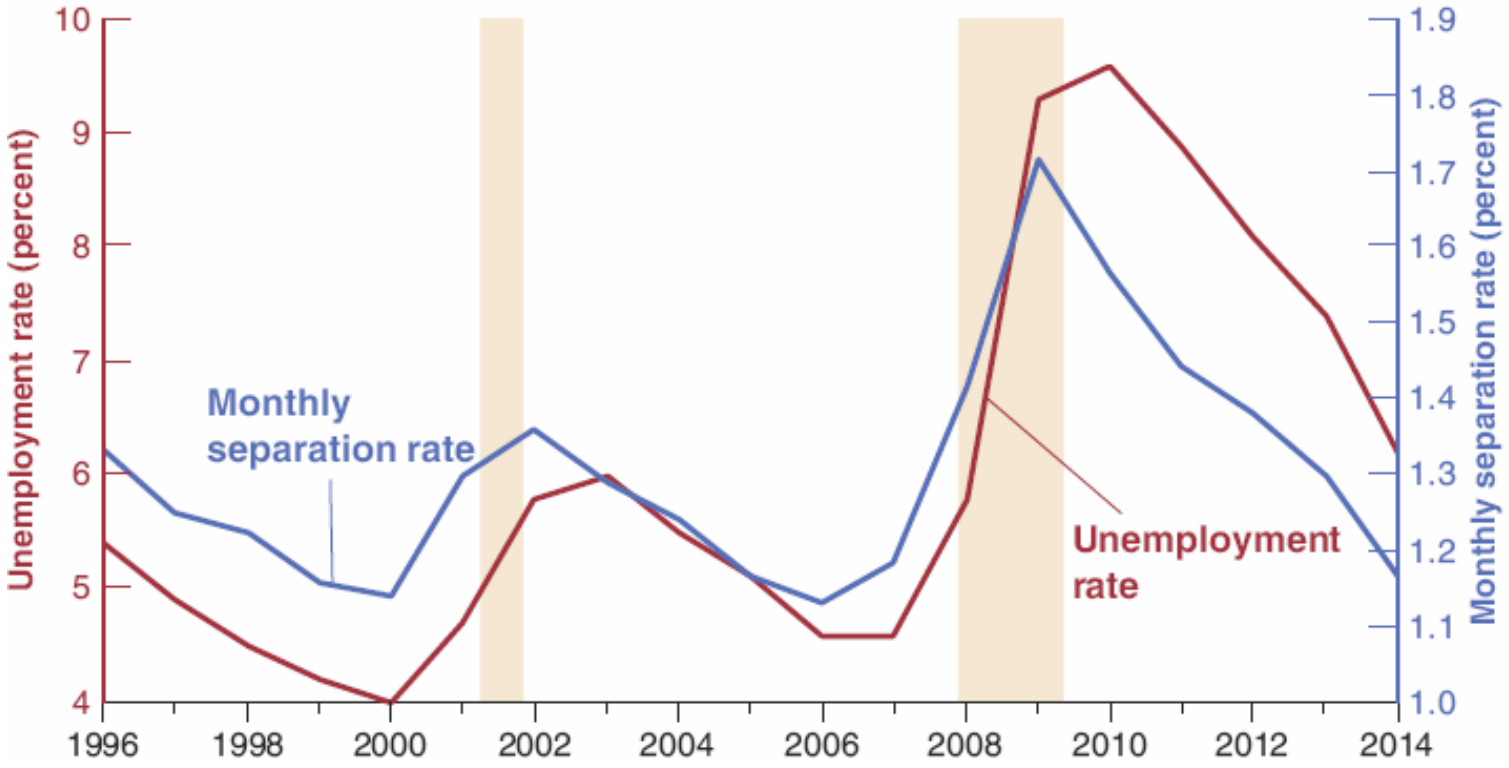


Sources: Series UNRATE: Federal Reserve Economic Data (FRED) http://research.stlouisfed.org/fred2/. Series constructed by Fleischman and Fallick, http://www.federalreserve.gov/econresdata/researchdata/.

When unemployment is higher, the proportion of unemployed finding jobs within one month is lower. Note that the scale on the right is an inverse scale.

2. Movements in Unemployment

Figure 7-5 The Unemployment Rate and the Monthly Separation Rate from Employment, 1996-2014



Sources: Series UNRATE: Federal Reserve Economic Data (FRED) http://research.stlouisfed.org/fred2/.
Series constructed by Fleischman and Fallick, http://www.federalreserve.gov/econresdata/research data/feds200434.xls.

When unemployment is higher, a higher proportion of workers lose their jobs.

4. Wage Determination

Given differences across workers and countries,
can we hope to formulate a **general theory of wage determination**?

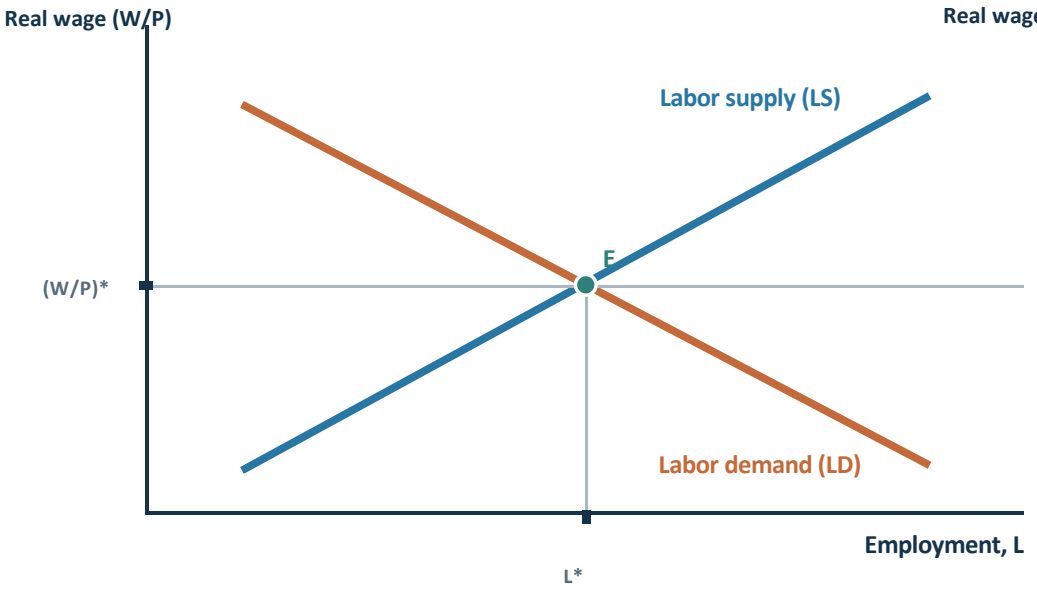
YES.

Although institutional differences influence wage determination
there are common forces.

Two labor-market equilibrium models

1. Competitive labor market

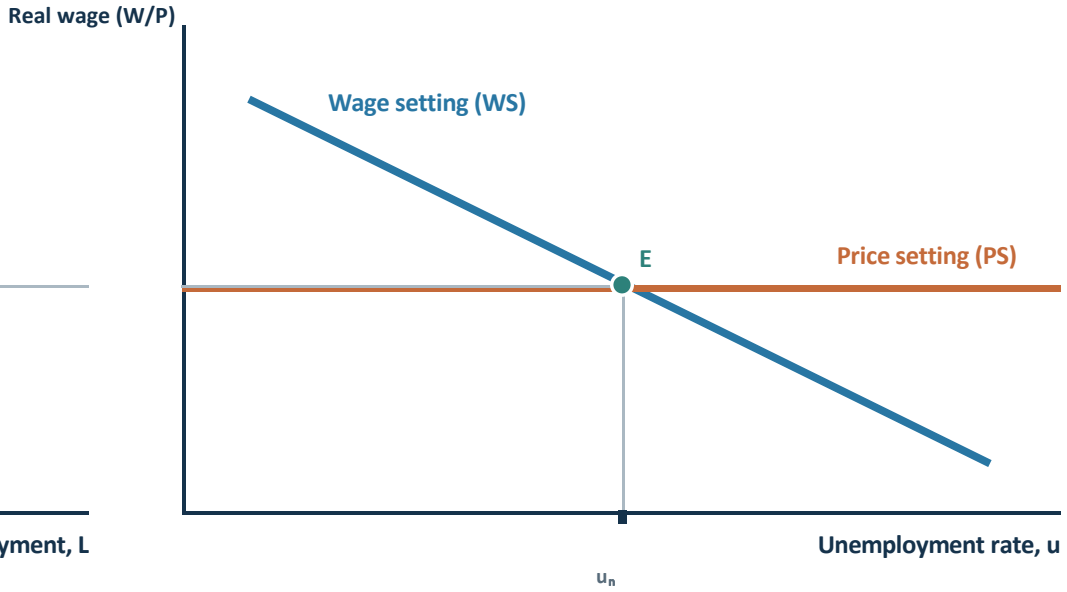
Real wage against employment



Equilibrium: LS = LD. The real wage clears excess labor supply or demand.

2. Wage-setting / price-setting

Real wage against unemployment



Equilibrium: WS = PS. Equilibrium unemployment can remain positive.

4. Wage Determination

- Sometimes wages are set by **collective bargaining**: a bargaining between trade unions and firms.
- Slightly more than 10% of U.S. workers' wages are set by collective bargaining (BLS).
- In Spain, **92.1% of employees eligible for collective bargaining were covered by a collective agreement in 2024** (OECD).
- Collective bargaining plays an important role in Japan and most European countries.
- The higher the skills needed to do the job, the more likely there is to be bargaining between employers and individual employees.

4. Wage Determination

- Workers are typically paid a wage exceeding their **reservation wage**: the wage that would make them indifferent between working or being unemployed.

- Wages typically depend on **labor-market conditions**: The lower the unemployment rate, the higher the wages.

- Given these two facts, economists have focused on two broad lines of explanation:

4. Wage Determination

1. Workers' **bargaining power** depends on:

   - How costly is for the firm to find other workers: **NATURE OF JOB**
   - How hard it is for workers to find another job if they were to leave the firm: **LABOR MARKET CONDITION**

P. Diamond, D. Mortensen and C. Pissarides received the 2010 Nobel Prize in Economics for working out the characteristics of a labor market with large flows and wage bargaining. https://www.nobelprize.org/prizes/economic-sciences/2010/prize-announcement/

4. Wage Determination

2. Efficiency wage theories link the productivity of the efficiency of workers to the wage they are paid.

- Firms may want to pay a wage above the reservation wage in order to decrease workers' turnover and increase productivity.
- Firms that see employee morale and commitment as essential to the quality of workers' work will pay more than those whose activities are routine.
- When unemployment is low, firms that want to avoid an increase in quits will increase wages to induce workers to stay with the firms.

4. Wage Determination

Example: Henry Ford and Efficiency Wages

Table 1 Annual Turnover and Layoff Rates (%) at Ford, 1913–1915

	1913	1914	1915
Turnover rate	370	54	16
Layoff rate	62	7	0.1

Source: Dan Raff and Lawrence Summers, "Did Henry Ford Pay Efficiency Wages?" *Journal of Labor Economics* 1987 5 (No. 4 Part 2): pp. S57–S87.

- In 1914, Henry Ford, the builder of Model-T, announced that his company would pay all qualified employees a minimum of $5.00 a day for an eight-hour day, compared to previously an average $2.30 for a nine-hour day.
- **The turnover rate (ratio of separation to employment)** plunged from 370% in 1913 to 16% in 1915: 31% of workers left each month.
- The layoff rate collapsed from 62% to nearly 0%.
- The average rate of absenteeism ran close to 10% in 1913 and decreased to 2.5% in 1914